

FR-018 M0 — Opportunity Discovery & Acquisition Engineering Spike

FR018 — `sources/optional/m0_spike.md`

Masterclass source package snapshot

Functional Requirement	FR-018
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M0 spike)** — **Accepted** by owner; unlocked M1;
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via `python scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

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FR-018 M0 — Opportunity Discovery & Acquisition Engineering Spike

Status: Complete (M0 spike) — **Accepted** by owner; unlocked M1; **GO to M1 under narrow scope** (URL-first executable path); M1 later redefined by owner as contracts-only — see

[fr018 m1 discovery contracts.md](#)

Date: 2026-08-07

Phase: Horizon 1B (lead FR)

Preceding: Horizon 1A complete/frozen (FR-008–FR-017); roadmap remap [changelog § 1.115](#); FR-008 acquisition boundary ([acceptance](#); [ADR-003](#)); FR-009 Opportunity SoT / duplicates ([acceptance](#); [ADR-004](#)); FR-016 Job Discovery compatibility placeholder ([fr016 m0 §23](#))

Scope (M0): Document-only architecture spike. **No production code.**

Does not begin (M0): M1 adapters, live board fetches, scheduled jobs, Opportunity mutation, Playwright, FR-008–017 redesign, Masterclass generation.

Owner direction accepted for this spike:

1. Mission is **lawful opportunity inflow** into the frozen Horizon 1A pipeline.
2. **Discovery Ingress** is the approved coordination direction — pressure-tested here.
3. Prefer APIs / feeds / email / exports / URLs over browser automation; **no scrape theatre**.

1. Executive summary

Horizon 1A delivered a complete application OS. Acquisition today remains **paste** and **local export** only. The largest remaining product bottleneck is getting suitable advertisements into that path with less manual effort.

Finding: The frozen `AcquisitionAdapter` → `ApplicationWorkflowRunner.start` → persist-before-review path is sufficient. FR-018 must **extend** that boundary, not redesign FR-008–017.

Finding: Discovery Ingress is valid **only** as a thin coordinator: resolve Opportunity Sources → instantiate adapters → invoke the existing runner. It must not become a second orchestration engine, Opportunity store, or business service.

Finding: Among lawful first milestones, **owner-supplied URL acquisition** is the best M1. Email alerts likely deliver higher *volume* later but need auth, MIME fixtures, and parser design that M0 cannot safely start in code. Official public APIs for major AU boards are generally unavailable. Playwright remains **last resort / NO-GO for M1**.

M0 recommendation: GO to M1 under narrow scope (URL adapter + thin owner CLI + fixture tests + ingress idempotency check against existing Opportunities). **NO-GO** on scrape-first, Playwright-default, parallel discovery catalogues, and agent discovery tools.

2. Engineering problem (restated)

How do we **lawfully discover and acquire** suitable job advertisements into CIC such that:

1. each posting enters with **explicit provenance**;
2. handoff uses the **frozen** FR-008/009 path without reopening Horizon 1A;
3. duplicates remain **linked, never merged** via existing identity facets;
4. the owner still **decides** apply / skip / defer;
5. we do not invent crawlers, a second SoT, or automation theatre?

This is **ingress engineering**, not application automation.

3. Dual-value and commercial honesty

Test	Result for narrow FR-018 (URL-first M1)
Improve interview/offer odds?	Yes, indirect — more suitable Opportunities enter review/submit
Reduce job-search effort?	Yes — removes paste/export for URL-available ads
Required infrastructure?	Modest — one adapter + thin ingress + CLI

Near-term commercial value: high relative to Recruiter Intelligence first (already remapped).

Learning value: high if architecture stays adapter-based and compliance-aware.

If M1 cannot fetch owner URLs fail-closed with fixtures, dual-value collapses — fail closed rather than scrape.

4. Current architecture inventory (reuse proof)

4.1 Ownership matrix (normative)

Concern	Owner FR	FR-018 role
Thin workflow runner, checkpoints, retries	FR-008	Reuse — call <code>start(adapter)</code> only
<code>AcquisitionAdapter</code> / <code>AcquisitionResult</code>	FR-008	Extend with new adapter implementations
Opportunity SoT, review queue, ranking	FR-009	Reuse — no second catalogue
Identity facets / duplicate detection	FR-009	Reuse — improve facets via <code>source_url</code>
Truth / BOPA / DOS / derive-only eval	FR-014–017	None — do not reopen
Discovery Ingress (thin)	FR-018 (proposed)	Resolve sources → adapters → runner
Scrape farms / crawlers	—	Out of scope

4.2 Existing contracts (do not widen for batch)

From `career_intelligence.orchestration.acquisition`:

Type	Role
<code>AcquisitionResult</code>	One posting + provenance (<code>source_kind</code> , <code>raw_content</code> , <code>posting</code> , optional <code>source_identifier</code> / <code>source_url</code> / title / company / <code>warnings</code> / <code>acquired_at</code>)
<code>AcquisitionAdapter</code>	Protocol: <code>source_kind</code> + <code>acquire()</code> -> <code>AcquisitionResult</code>
<code>AcquisitionError</code>	Fail-closed acquisition failure

Implemented today: `PasteAcquisitionAdapter` , `LocalFileAcquisitionAdapter` .

Reserved literals (unimplemented): `url` , `api` , `email` , `saved_search` , `playwright` , `other` .

Runner: `ApplicationWorkflowRunner.start(source: AcquisitionAdapter | PasteJobInput)` — does **not** branch on `source_kind` .

4.3 Handoff sequence (frozen)

```
acquire → validate_normalise → analyse → assess → match → strategy
      → persist (decision = None) → owner_review → record_decision
```

Matches FR-016 M0 §23 future Job Discovery compatibility placeholder.

5. Opportunity Source vs Opportunity (domain)

Concept	Durable?	Role
Opportunity	Yes — data/ opportunities/ SoT	Analysed candidate; review metadata; pipeline
OpportunitySource	No — transient ingress metadata	Channel + locator + auth posture for one acquire attempt
Discovery run / audit (optional later)	Recovery only	Must not catalogue business Opportunities

Why Opportunity remains the only durable business record

1. ADR-004: single SoT; review queue and duplicates are **derived**.
2. A “seen jobs” discovery store would recreate the pre-FR-009 checkpoint-as-catalogue failure mode.
3. Skip/defer/apply auditability requires Opportunities created through the analysis path (or an explicit owner-approved exception — none proposed here).

OpportunitySource may exist as a typed value object or CLI argument shape in M1; it must dissolve into `AcquisitionResult` provenance once acquire succeeds.

Interview takeaway: Ingress metadata is ephemeral; business truth is the Opportunity aggregate.

6. Discovery Ingress — pressure test

6.1 Approved responsibility (normative)

Discovery Ingress **may only**:

1. Resolve Opportunity Sources (locators → adapter instances).
2. Instantiate the correct `AcquisitionAdapter`.
3. Invoke existing `ApplicationWorkflowRunner.start(adapter)` (or equivalent public entry that already accepts adapters).
4. Optionally **pre-check** existing Opportunities for definite identity matches (idempotency warn/skip) — read-only against Opportunity service.
5. Surface warnings / counts to the owner CLI.

6.2 Forbidden responsibilities (normative NO-GO)

Discovery Ingress **must not**:

Anti-pattern	Why rejected
Second workflow / orchestration engine	Reopens ADR-003; duplicates FR-008
Persist Opportunities itself	Bypasses analyse/assess/strategy trust
Own a durable “candidate jobs” SoT	Second catalogue; ADR-004 violation
Rank, assess, package, truth-validate	Owned by frozen FRs
Submit, pipeline advance, recruiter contact	Human gates + later FRs
Batch-mutate ToolPolicy / agent allow-lists	ADR-007/008 forbid discovery as agent tools

6.3 Alternatives considered

Alternative	Verdict	Why
A. Adapters only; no ingress type	Weaker	CLI would re-implement resolution/idempotency ad hoc
B. Fat Discovery service (analyse inside)	Reject	Redesigns Horizon 1A
C. Multi-agent Acquisition Specialist (MVP)	Reject	Theatre; FR-016 placeholder only
D. Thin Discovery Ingress	Accept	Matches adapter isolation + reuse

Principle demonstrated: Orchestration coordinates; adapters channel; services execute — Ingress is coordination only.

Interview takeaway: “Thin ingress, fat reuse” — the smallest new component that preserves a frozen pipeline.

7. Acquisition mechanism comparison

Representative mechanisms (owner workflow is one validation example, not the only design input).

Mechanism	Complexity	Stability	Maintenance	Lawfulness	Platform dependence	Owner UX	Commercial value	Learn value
Official APIs	Low–med if exist	High	Low	Best	Provider	Excellent	High	Med
Structured feeds / RSS	Low–med	Med	Low	Usually OK	Provider	Good	Med–high	Med
Email alerts	Med	Med	Med–high (parsers)	Owner mailbox	Provider templates	Strong volume	High	High
Manual URLs	Low	High	Low	Owner-supplied fetch	Board HTML risk	Immediate	High for M1	High (facets)
Export files	Low	High	Low	High	None	Manual	Med (exists)	Low
Supported integrations	Unknown	Unknown	Med	Depends	High	Good	Unknown	Med
Browser automation	High	Brittle	High	ToS / anti-bot risk	Extreme	Fragile	False economy	Low if scrape story

7.1 URL-first hypothesis — pressure test

Hypothesis: URL acquisition should be M1.

Challenge — email-first?

Email alerts often deliver the highest *passive* volume for SEEK/LinkedIn-style searches. However M1 email requires: mailbox access posture, MIME/HTML fixtures, template drift handling, and security review. That is a larger first slice and risks shipping a brittle parser as the “hello world.”

Challenge — API-first?

Desk review (spike-level): major AU/consumer job boards used in this search do **not** expose a simple owner-usable public “list suitable AI jobs” API for CIC. Building against unavailable APIs delays value.

Challenge — export-only improvements?

Already shipped; marginal dual-value vs URL.

Challenge — Playwright-first?

Technically tempting; commercially and legally weak; contradicts principles failure mode #2 and FR-008 avoid-list.

Verdict: Confirm URL-first for M1. Email/feeds are the primary M2 hypothesis after URL proves handoff + identity facets. Playwright remains out of M1–M2 unless a later spike revises with explicit owner approval.

Why URL wins as first milestone

1. Lawful owner-supplied locator.
2. Immediately improves `derive_source_facets` (Seek/LinkedIn/Indeed IDs).
3. Reuses reserved `source_kind="url"`.
4. Smallest fail-closed adapter (fetch → text → `AcquisitionResult`).
5. Teaches provenance + idempotency without inbox complexity.

8. Provenance design (validated)

8.1 Required fields on every successful acquire

Field	Required	Notes
<code>source_kind</code>	Yes	e.g. <code>url</code> for M1
<code>raw_content</code>	Yes	Non-empty; fail closed otherwise
<code>posting</code>	Yes	Typed <code>JobPosting</code>
<code>source_identifier</code>	Strongly required for M1	Canonical URL string or stable hash of URL
<code>source_url</code>	Strongly required for M1	Feeds identity facets
<code>acquired_at</code>	Yes	Ingress timestamp
<code>warnings</code>	As needed	Soft issues; not silent success on empty body
title / company	Optional	Prefer extracted/grounded; never invent

8.2 Duplicate detection compatibility

Existing FR-009 behaviour (reuse):

- Definite: same `canonical_url` / `source_url` / (`platform` + `platform_job_id`)
- Probable / possible: company/title/location/fingerprint clusters
- Fingerprint alone is **not** a merge key (ADR-004)

URL-first M1 maximises definite matches. Paste-without-URL remains weaker — not the FR-018 focus.

8.3 Idempotent acquisition behaviour (normative for M1+)

Before `runner.start` for a URL (or other locator with definite facets):

1. Query Opportunity service for definite identity match.
2. If match: **warn/skip** (or owner `--force` only if later approved) — **never merge/delete**.

3. If no match: proceed through full FR-008/009 path.
4. Post-persist: existing duplicate detection may still propose probable groups; owner confirms.

Ingress idempotency is **advisory skip**, not silent data destruction.

9. Pipeline integration (no Horizon 1A redesign)

```

OpportunitySource (transient)
  → Discovery Ingress
  → AcquisitionAdapter.acquire() → AcquisitionResult
  → ApplicationWorkflowRunner.start(adapter)
  → frozen PRE_APPROVAL_SEQUENCE
  → Opportunity SoT + derived review queue

```

Frozen FR	Change required?
FR-008 runner / checkpoints / retries	No
FR-009 SoT / queue / dupes / rank	No (read for idempotency)
FR-014 truth	No
FR-015 BOPA	No — no discovery tools
FR-016 DOS/OBS	No — placeholder hand-in remains future
FR-017 eval	No

Optional later ADR (M1+): record Discovery Ingress boundary decisions if they exceed what ADR-003 already implies for adapters.

10. Human approval boundaries

Action	Allowed in FR-018
Owner supplies URL / later email export	Yes
Fetch content for owner-supplied locator	Yes (M1+)
Create Opportunity via existing persist-before-review	Yes
Owner apply / skip / defer	Required (unchanged)
Silent submit / outreach / pipeline advance	No
Agent-initiated mutating discovery	No without new ADR + owner acceptance

11. Risks and mitigations

Risk class	Risk	Mitigation
Technical	HTML layout drift on URL fetch	Extract visible text conservatively; warnings; fixtures; fail closed on empty
Technical	Dual SoT temptation	Normative: no discovery catalogue
Maintenance	Email parser sprawl (M2)	Fixture corpus; narrow providers; defer if unstable
Platform	Rate limits	Owner-paced CLI first; no aggressive schedulers in M1
ToS	Treating fetch as scrape farm	Owner-supplied URLs only in M1; no mass crawl; document posture
Owner UX	Fetch fails on JS-heavy pages	Warning + fallback to paste/export; Playwright only if later GO
Commercial	Building theatre instead of inflow	Dual-value gate; URL M1 ships real path
Portfolio	"We scraped LinkedIn" narrative	Refuse; tell adapter/compliance story

12. Recommended scope (post-GO)

In scope (FR-018)

- Thin Discovery Ingress (resolve → adapter → runner)
- New lawful adapters starting with **URL**
- Provenance-complete results
- Ingress idempotency against Opportunity identity
- Thin owner CLI
- Fixture-based automated tests (no network in CI)

Out of scope / NO-GO slices

- Scrape-first / uncontrolled crawlers / access-control bypass
- Playwright as M1 default
- Parallel Opportunity catalogue
- Redesign of FR-008–017
- Auto apply / submit / recruiter messaging
- Discovery on BOPA/DOS allow-lists

- Dashboards; Academy artefacts during engineering

Deferred

- Email alert ingress (M2 hypothesis)
- Feeds / saved-search (M2–M3)
- Official APIs if/when available
- Scheduling / multi-URL batch UX polish (M3)
- Playwright last-resort spike (only with owner + revised GO)

Proposed milestones (planning; M1 not started)

Milestone	Intent
M0	This spike — Complete pending owner accept
M1	<code>UrlAcquisitionAdapter</code> + thin CLI + fixtures + idempotency pre-check
M2	Email and/or feed ingress (M0-ranked)
M3	Batch/ops hardening
M4	Evaluation / acceptance freeze

13. Testing posture (for M1 planning; not executed in M0)

- Deterministic fixtures; **no network in CI**
- Fake HTTP / saved HTML for URL adapter unit tests
- Functional: adapter → runner → identity facets on fixtures
- Manual owner validation on real saved URLs after M1

14. Learning and interview value

Major decisions (teachable)

D1 — Thin Discovery Ingress

- **Why:** Coordinate without redesigning the runner.
- **Alternative:** Fat discovery orchestrator.
- **Rejected:** Reopens frozen Horizon 1A.
- **Principle:** Reuse before redesign; adapters channel.
- **Interview:** Smallest boundary that preserves SoT ownership.

D2 — Keep `AcquisitionAdapter` **one-posting**

- **Why:** Runner is source-agnostic; batch belongs at ingress.
- **Alternative:** Widen Protocol to `acquire_many`.
- **Rejected:** Forces runner/API churn.
- **Principle:** Deterministic first; narrow interfaces.
- **Interview:** Don't break frozen contracts for convenience.

D3 — URL-first M1

- **Why:** Lawful, facet-rich, smallest slice.
- **Alternative:** Email-first or Playwright-first.
- **Rejected (for M1):** Complexity / ToS.
- **Principle:** Validate first; no scrape theatre.
- **Interview:** Compliance-aware integration beats clever scraping.

D4 — Opportunity-only SoT

- **Why:** ADR-004 lesson.
- **Alternative:** Discovery candidate DB.
- **Rejected:** Dual catalogue.
- **Principle:** One system of record.
- **Interview:** Derived views over durable records, not shadow stores.

Academy evidence left by this spike

Alternatives, invariants, dual-value honesty, GO/NO-GO table, milestone non-goals, provenance/ idempotency rules — sufficient for later Acceptance → Masterclass → Interview Brief/Deck after freeze.

No Academy artefacts generated now.

15. Go / no-go table

Slice	Verdict
Thin Discovery Ingress (resolve → adapter → runner only)	GO
Keep AcquisitionAdapter narrow (one posting)	GO
Opportunity as sole durable SoT; Source transient	GO
URL-first M1	GO
Email/feeds as M2+	GO (deferred)
Official APIs if unavailable	DEFER until real API exists
Playwright / scrape-first	NO-GO for M1–M2 default
Parallel discovery Opportunity catalogue	NO-GO
Redesign FR-008–017	NO-GO
Agent discovery tools on BOPA/DOS	NO-GO
Auto apply / outreach	NO-GO
M1 implementation in this spike	NO-GO (docs only)

16. M0 recommendation

GO to M1 under narrow scope.

1. Owner accepts this M0 spike.
2. M1 plans/implements **URL acquisition** + thin Discovery Ingress + CLI + fixture tests + idempotency pre-check.
3. Do **not** begin Playwright, email, or APIs in M1 unless owner revises to **GO WITH REVISED SCOPE**.
4. Recruiter Intelligence remains **FR-019**.

Alternate owner replies

Choice	Meaning
Accept M0 as written	Unlock M1 planning/implementation under URL-first
Accept with revisions	List changes (e.g. email-first)
Defer FR-018	Capacity on live applications; record in changelog
Reject	Architecture not justified; close or redesign spike

No M1 work begins without an explicit choice above.

17. M0 definition of done

- [x] Engineering problem restated
 - [x] Dual-value honesty recorded
 - [x] Architecture inventory / reuse matrix
 - [x] Discovery Ingress pressure-tested (allowed vs forbidden)
 - [x] Mechanism comparison across representative sources
 - [x] URL-first hypothesis challenged and confirmed
 - [x] Opportunity vs OpportunitySource clarified
 - [x] Provenance + dedup + idempotency specified
 - [x] Pipeline integration without Horizon 1A redesign
 - [x] Risks + mitigations
 - [x] Learning / interview decision records
 - [x] Explicit GO / NO-GO table
 - [x] Owner next-step block
 - [x] No production code / adapters / runtime changes
-

18. Final repository status (at M0 close)

Item	Status
Horizon 1A FR-008–FR-017	Frozen / unchanged
FR-018 M0 spike	Complete — GO to M1 under narrow scope
M1 URL adapter / CLI	Not started — awaits owner acceptance
Production acquisition beyond paste/export	Unchanged

M0 COMPLETE

GO to M1 under narrow scope

FR-018 READY FOR M1 PLANNING (IF OWNER ACCEPTS)